

A 10-b 50-MS/s 820- μ W SAR ADC With On-Chip Digital Calibration

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Abstract—This 10-b 50-MSamples/s SAR analog-to-digital converter (ADC) features on-chip digital calibration techniques, comparator offset cancellation, a capacitor digital-to-analog converter (CDAC) linearity calibration, and internal clock control to compensate for PVT variations. A split-CDAC reduces the exponential increase in the number of unit capacitors needed and enables the input load capacitance to be as small as the kT/C noise restriction. The prototype fabricated in 65 nm 1P7M complementary metal–oxide semiconductor with MIM capacitor achieves 56.6 dB SNDR at 50-MSamples/s, 25-MHz input frequency and consumes 820 μ W from a 1.0-V supply, including the digital calibration circuits. The figure of merit was 29.7 fJ/conversion-step under the Nyquist condition. The ADC occupied an active area of 0.039 mm².

Index Terms—Analog-to-digital converter (ADC), digital calibration, successive approximation register.

I. INTRODUCTION

A SYSTEM-ON-CHIP (SoC) for ultrasound systems and sensors needs two or more analog-to-digital converters (ADCs) for multichannel inputs. Power and area reductions are very important since they enable more channels to be implemented in one chip. Reduction of the number of chips results in a miniaturization of the system. Technology scaling improves the digital performance of SoC, enabling large-scale on-chip integration and high-speed operation with superior energy efficiency. On the other hand, the poor performance of the metal–oxide semiconductor field-effect transistors (MOSFETs) in the saturation region makes it difficult to design a high gain amplifier, and the lower power-supply voltage intensifies floating switch and thermal noise problems. As a result, it is steadily becoming more difficult to design pipelined ADCs because the ADC's performance depends on the amplifier's performance. In addition, at a lower supply voltage, the analog signal suffers from the increased resistance of the analog switches. Successive approximation register (SAR) ADCs are more compatible with technology scaling because they do not need operational amplifiers and have fewer floating switches. The only analog part is the comparator, whose design is close to that of a digital regenerative latch. This means that

no static current flows if charge leakage from the capacitor digital-to-analog converter (CDAC) can be avoided; hence, the power consumption will be a linear function of the conversion rate. In fact, a low-power pipelined ADC that does not require amplifiers has been developed [1]. However, the pipeline architecture has more circuits than the SAR architecture. Hence, an SAR architecture would still be a better choice for low-power and compact ADCs.

Designing the ADC using small devices takes advantage of the scaling merits. Higher conversion rate and smaller area are direct benefits. Smaller area in turn contributes to reducing the power consumption, because the smaller parasitics reduce the required drive power. However, accuracy suffers from device mismatch. Many digitally assisted techniques have been reported in an attempt to resolve this issue. Foreground calibration was reported [2] as a way to align the mismatch of the capacitors, but the actual calibration was done via off-chip software processing. A nonbinary CDAC design based on unit capacitors [3] has redundancy for correcting the misjudgments of the comparator and/or incomplete CDAC settling. However, the linearity is still affected by capacitor mismatch. Nonbinary designs using C-2C CDACs are reported in [4]. Redundancy works because the upper bit decision can have a larger redundancy. However, this structure needs additional stages as well as digital postprocessing. Another nonbinary design Radix = 1.86 was implemented with individually arranged capacitors which can maintain linearity by using the perturbation technique [5]. This technique can correct the mismatch of the radix between each bit. However, it also needs additional stages and digital postprocessing.

This paper presents on-chip digital calibration techniques to correct the CDAC mismatches based on [2] and an internal clock control that enables a very slow conversion rate while avoiding the charge leakage problem of the CDAC. Section II describes the design and architecture of the proposed SAR ADC. Section III presents the calibration techniques and Section IV shows the experimental results. Conclusions are given in Section V.

II. CIRCUIT DESIGN

A. ADC Architecture

Fig. 1 shows the block diagram of the SAR ADC. The actual design is differential, but the figure shows a single-ended design for simplicity. It consists of a 10-b split-CDAC, a comparator, an internal clock generator, SAR controller, and a digital calibration circuit. The split-CDAC can reduce the number of unit capacitors relative to a conventional binary weighted CDAC design; however, it is very sensitive to mismatch as well as to parasitic capacitance. This causes a linearity error as a result. A vari-

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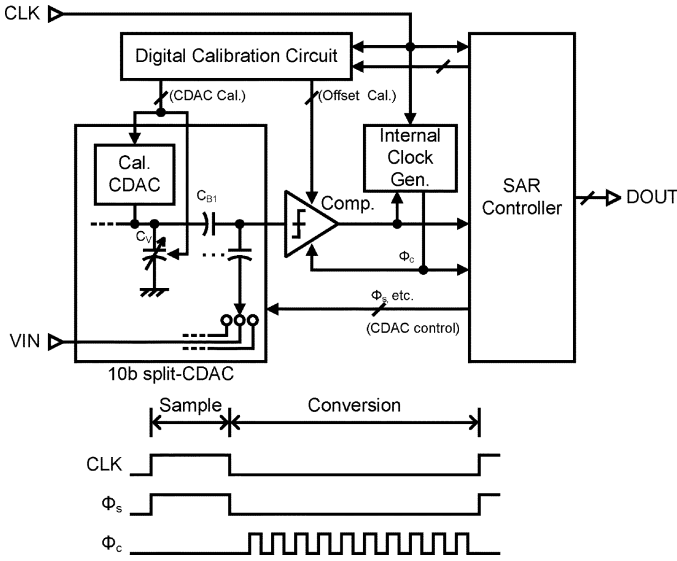


Fig. 1. ADC architecture.

able capacitor C_V is used to compensate for the mismatches of the splitting capacitor C_{B1} . The capacitance of C_V is adjusted prior to the conversion. The calibration CDAC corrects the mismatches of the upper three bits by operating in parallel with the main CDAC. The comparator offset is cancelled in order to correctly detect the mismatches of the CDAC.

The SAR ADC samples the analog signal during the CLK = "H" period and completes ten comparisons for 10-b conversion on the basis of an internal clock Φ_c , which is generated by a loop consisting of the same comparator used with the CDAC and delay elements. This loop is dependent on the PVT conditions, so its frequency is controlled during the conversion. Section III describes the CDAC linearity calibration and internal clock generation.

B. CDAC

The CDAC is a split capacitor array designed with the upper 4 b ($C_9 \sim C_6$), middle 4 b ($C_5 \sim C_2$) and lower 2 b ($C_1 \sim C_0$) arrays as shown in Fig. 2. C_X is used to reduce the input load capacitance. Its capacitance corresponds to the equivalent capacitance of the L-side. The analog signal is sampled by C_X ; as a result, the capacitors on the L-side do not need to sample the analog signal. C_H and C_L attenuate the power-supply VDD to generate an adequate full-scale range of ADC. They are connected to the analog signal during the sample phase and are respectively switched to ground and VDD during the conversion phase. The common mode level is designed to be at $V_{DD}/2$, so C_H and C_L have the same capacitance value. The capacitance is designed to be $7C$, where the unit C is designed to be 17 fF, so that the full-scale range is around VDD [peak-to-peak differential].

The total amount of input load capacitance is 30 C . This means that the input load capacitance is 510 fF, which is large enough to meet the kT/C noise requirement for covering process variations. Mismatches of the CDAC are compensated by C_V and the calibration CDAC. The total number of unit

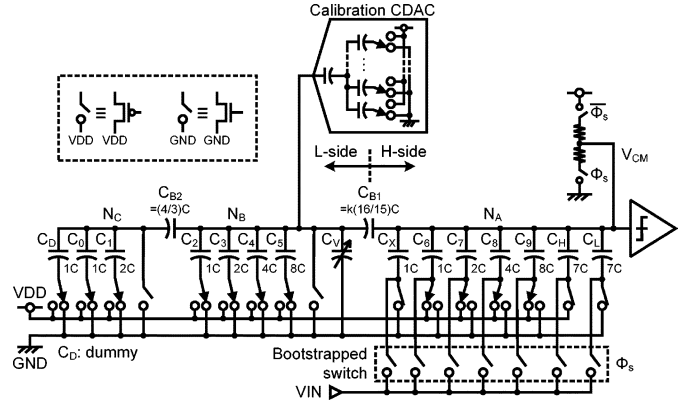


Fig. 2. The 10-b split-CDAC.

capacitors is 300 for both differential sides, which is approximately 1/7th the amount of a full straight binary weighted CDAC.

The switches in the CDAC are connected to either VDD or ground, so no complementary switch is needed. They are designed by using either NMOS or PMOS to save the area and power. Bootstrapped switches based on [6] are used for analog signal sampling to suppress distortion.

C. Comparator and Internal Clock Generator

Fig. 3 shows the internal clock generator, which includes a comparator in its loop. A strong arm latch type comparator without a preamplifier (Fig. 5) is used. It does not consume static current. Both of the comparator outputs, V_{QP} and V_{QM} , become "H" during the sample phase. During the conversion phase, one of them goes to "L", which enables the end of the comparison to be detected from the output signal V_A of the AND gate. We carefully designed the threshold of the AND in order to avoid incorrect detections resulting from the transient response of the comparator. V_B is generated by adding an adequate delay T_d to V_A . Finally, Φ_c goes to "H" and the comparator resets after completing one loop. This loop is repeated ten times during the conversion phase. The frequency of Φ_c depends on the comparator response delay and T_d . T_d is automatically controlled by Φ_c to be approximately 700 MHz in order to make ten comparisons in 15 ns. Five nanoseconds is the period given for the analog signal sampling at 50 MSamples/s.

III. DIGITAL CALIBRATION

A. Comparator Offset Cancellation

Fig. 4 shows the calibration scheme and status of each circuit after the ADC is powered on. The principle of the CDAC linearity calibration is to correct its nonlinearity by comparing the capacitance mismatches between the upper bit capacitor and the rest of the lower bit capacitors including the dummy. Therefore, as shown in Fig. 4, the comparator offset should be canceled prior to the CDAC linearity calibration; otherwise, it will result in a mismatch in the CDAC, causing a linearity error. Various comparator offset cancellation techniques have been published [7]–[9]. A dynamic offset control technique adjusts the

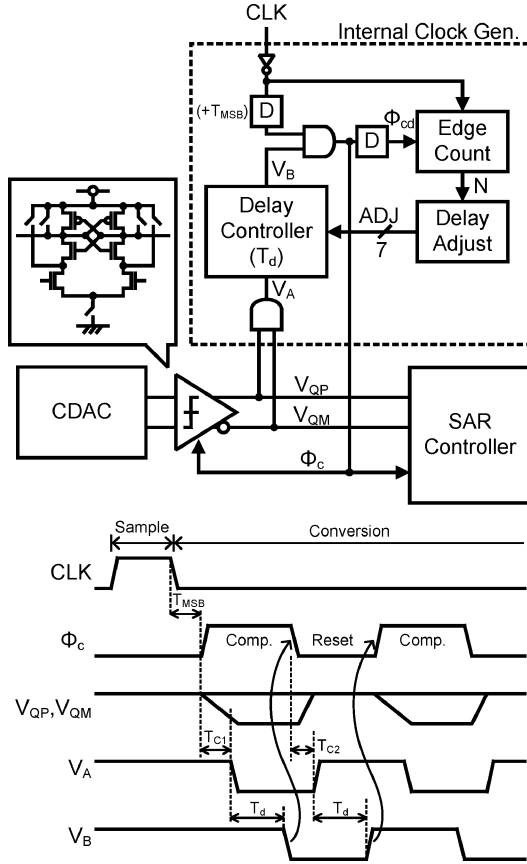


Fig. 3. Internal clock generator.

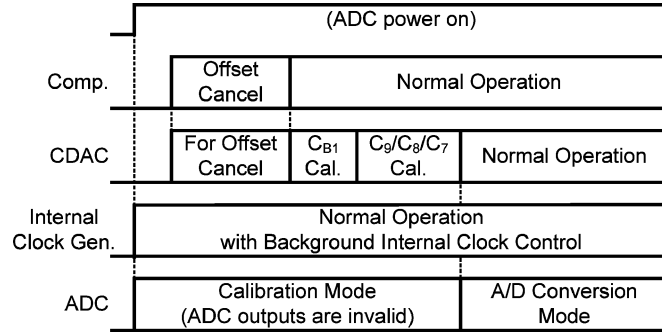


Fig. 4. Calibration scheme and status of each circuit.

output load capacitance controlling the offset; this, however, degrades the response [7]. Another dynamic offset cancel technique [8] controls the clock skew, which results in less of a response penalty; however, the timing control is sensitive to PVT variations. The threshold control technique shown in Fig. 5 can be used to cancel the comparator offset [9]. The threshold of either of the differential pair M_{IP} or M_{IM} is controlled by forward body biasing. As a result, the difference between I_P and I_M is minimized. For instance, if $I_P > I_M$, V_{BM} is raised to increase I_M . This technique does not have a response penalty and does not require sensitive timing control.

The offset cancellation is done by feeding the same analog input signal to the comparator by connecting V_{IP} and V_{IM} to

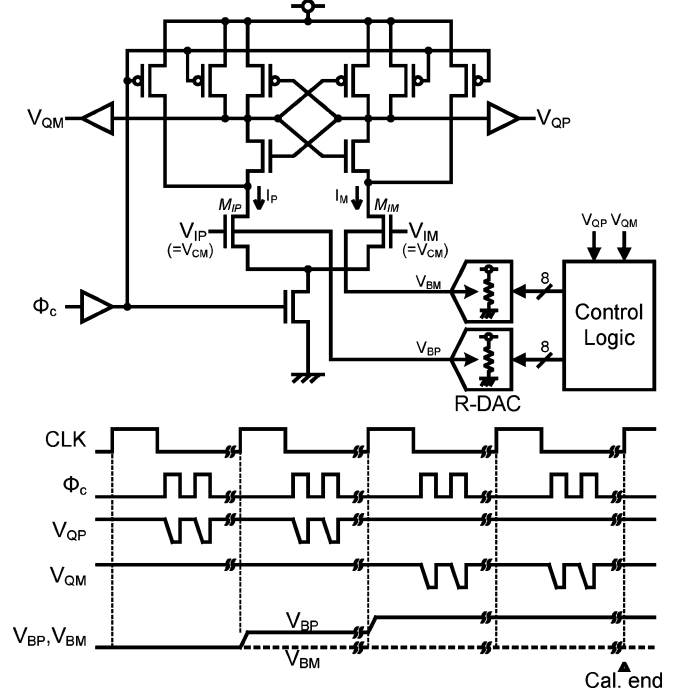
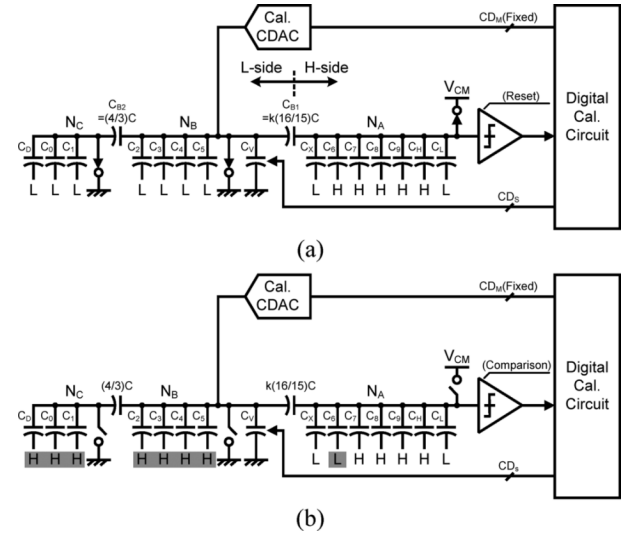
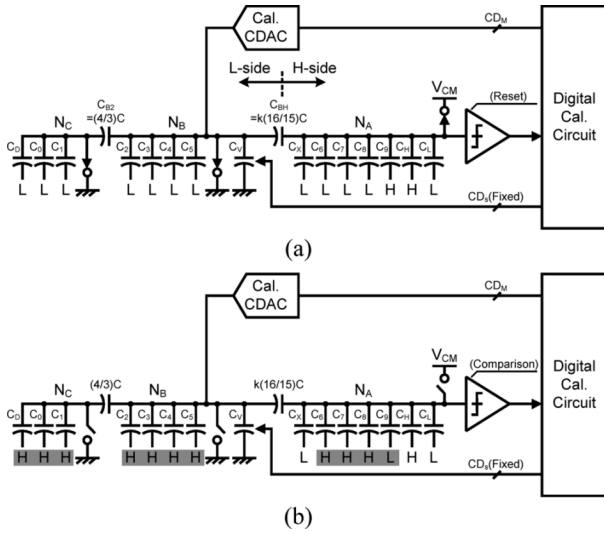


Fig. 5. Comparator with offset cancellation.

Fig. 6. C_{B1} calibration. (a) Sample phase. (b) Conversion phase.

the common level V_{CM} . V_{BP} and V_{BM} are initially connected to ground (0 V). Several comparisons are made to avoid thermal noise effects inherent in the comparator. The final result, “L” or “H,” is determined by the majority value and is stored in the register. If the result is the same as the previous one, either V_{BP} or V_{BM} is raised accordingly to reduce the offset voltage. This routine is continued until the result reverses. The comparator offset cancellation then finishes by fixing V_{BP} and V_{BM} .

The body biasing is controlled by a resistor DAC. This DAC consists of 256 taps and a polarity selector, amounting to a total of 512 steps. The minimum step and dynamic range are designed to cover the offset distribution with an error smaller than 0.25 LSB at the ADC output. The resistor DAC dissipates 20 μ A.

Fig. 7. C_9 calibration. (a) Sample phase. (b) Conversion phase.

B. CDAC Linearity Calibration

The CDAC linearity error caused by parasitics and process mismatch is calibrated after the comparator offset cancellation is completed. The mismatch as well as the parasitic effects of the split capacitor C_{B1} is calibrated by a variable capacitor C_V , and the mismatch of the upper three bits are corrected by the calibration CDAC. C_{B1} could be calibrated by using the calibration CDAC; however, the CDAC would then need a larger dynamic range, resulting in a larger area. Using C_V , however, obviates the need for a large dynamic range, so it is simpler than using the calibration CDAC. C_{B2} between the nodes N_B and N_C is not calibrated because its error is negligible.

C_{B1} is calibrated first because the upper three-bit ($C_9 \sim C_7$) calibration uses the lower bit capacitors. The principle of the C_{B1} calibration is shown in Fig. 6 [8]. In the ideal case with no parasitics, C_{B1} should be $16C/15$; in practice, however, its value is set to be $k(16C/15)$ ($k > 1$). The calibration CDAC's state is fixed during the calibration period of C_{B1} . In the first step of the calibration, C_V is given a minimum capacitance. In the sample phase, all of the capacitors on the L-side are connected to ground ("L") and all capacitors on the H-side are connected to VDD ("H"), biasing the comparator input node N_A to the V_{CM} [Fig. 6(a)]. Next, in the conversion phase, N_A is disconnected from V_{CM} and all of the L-side capacitors are switched to VDD ("H"), and C_6 is switched to ground ("L") [Fig. 6(b)]. Mismatches between all of the L-side capacitors and C_6 appear on N_A and are judged by the comparator. If C_V is smaller than the value needed to compensate for the mismatch of C_{B1} , the equivalent capacitance on the L-side will be larger than C_6 . The result is a higher voltage on N_A and the comparator judges the result as a "H." The result is fed to the digital calibration circuit, and a larger capacitance is chosen for C_V . This routine is repeated until the comparator's result becomes "L." The comparator's result is also based on selecting the majority decision; the same method is used for the comparator offset cancellation. The step of C_V is designed to be smaller than 0.25 LSB.

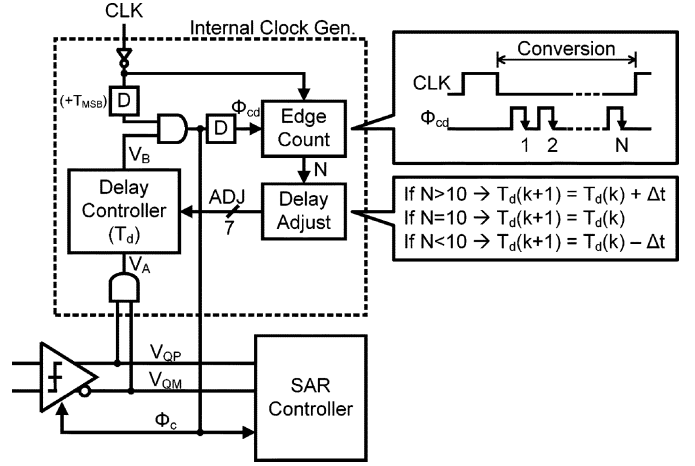
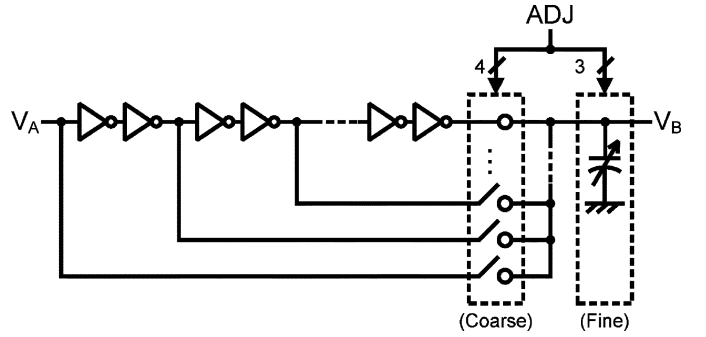
Fig. 8. T_d control implementation.

Fig. 9. Delay controller.

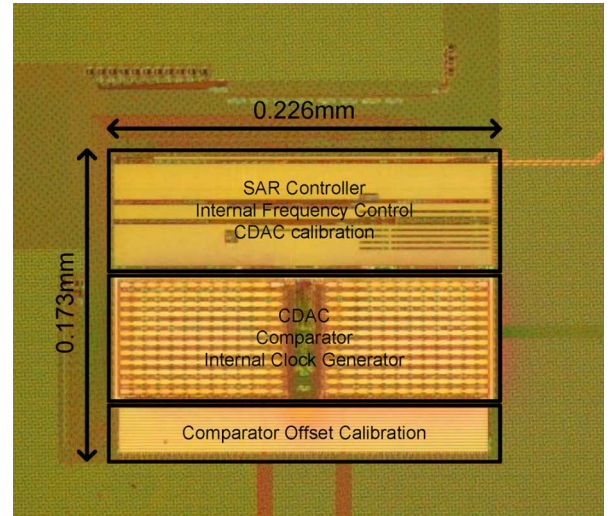


Fig. 10. Die photograph.

C_9 , C_8 , and C_7 are calibrated after C_{B1} . The calibration CDAC has the same structure as the switched capacitor DAC architecture in order to operate with C_9 , C_8 and C_7 . The calibration CDAC is connected to node N_B , not N_A [10]; therefore, it influences the main CDAC as a parasitic. However, it is compensated during the C_{B1} calibration. Fig. 7 shows the principle of the C_9 calibration, which is similar to that of C_{B1} . In the sample phase, C_9 is connected to VDD ("H") and the rest

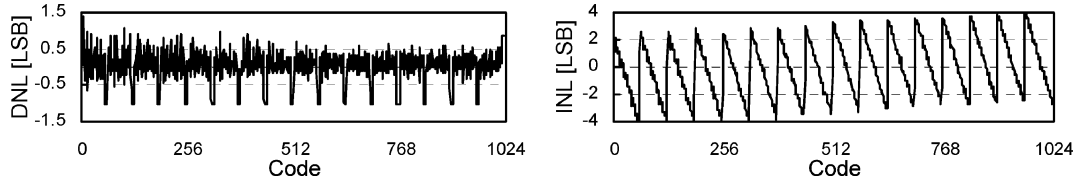


Fig. 11. Measured DNL and INL before calibration.

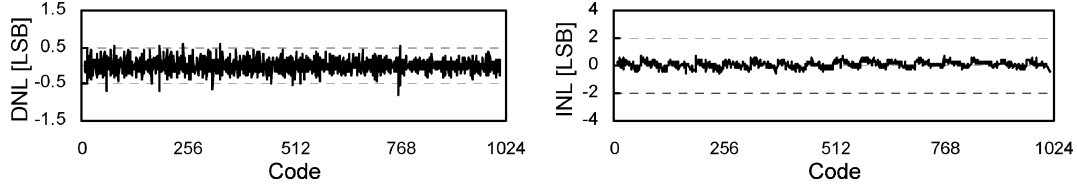


Fig. 12. Measured DNL and INL after calibration.

of the capacitors, C_8 to C_0 , dummy capacitor C_D and C_X are connected to ground (“L”), biasing the N_A to V_{CM} , and N_B and N_C to ground, respectively [Fig. 7(a)]. Next, in the conversion phase, N_A , N_B , and N_C are disconnected. C_9 is switched to ground (“L”), and C_0 to C_8 including C_D are switched to V_{DD} (“H”). C_X remains connected to ground (“L”) [Fig. 7(b)]. If there is no mismatch between C_9 and the rest of the capacitors, C_0 to C_8 and C_D , N_A stays at V_{CM} . In the other words, the mismatch causes some change in N_A . The change is detected by the comparator on the basis of the majority decision and is fed back to the calibration CDAC to cancel it. This is repeated as in the C_{B1} calibration, C_8 and C_7 are also calibrated in the same way [2]. The calibration CDAC control codes are stored and used during the actual conversion.

C. Internal Clock Control

The internal clock Φ_c is generated by a loop, which includes the comparator and a delay controller (Fig. 3). The cycle period is $T_{C1} + T_{C2} + 2T_d$. Thirty percent of the Φ_c period $T_{C1} + T_{C2}$ is spent for comparison. The remaining 70% $2T_d$ is spent for the CDAC response and SAR controller. It is useful to synchronize T_d with the CDAC and SAR controller. However, prior ADCs use a fixed T_d and provide no means to control it [4]. T_d has a large PVT variation, so it becomes hard to synchronize it with the CDAC and SAR controller. If T_d becomes shorter than the CDAC and SAR controller delay, an incomplete settling will degrade the accuracy. An excessively long T_d limits the number of available comparison cycles. The T_d controller implemented for this ADC is shown in Fig. 8. It delivers the longest T_d within the comparison period given by the external clock. The T_d controller works in the background, as shown in Fig. 4, because the voltage and temperature conditions are not constant during operation.

The algorithm is based on counting the Φ_c cycles N during the conversion phase; T_d is calibrated to make the number of cycles N equal to 10. In practice, a delayed Φ_c signal (Φ_{cd}) is actually used to provide a wide enough margin, and this results in an improvement in the clock jitter tolerance. The transition point of the comparator as it starts to reset is used for the trigger edge to count the Φ_c cycles. If $N > 10$ ($N < 10$), T_d is made longer (shorter) by using the ADJ code. In this way, T_d converges to

provide the longest comparison period. Φ_c stays at “L” during the sampling phase and starts to work after a delay of T_{MSB} after the falling edge of CLK. The delay T_{MSB} is designed to meet the CDAC settling time requirement for MSB decision, and it is generated by a fixed delay circuit.

Fig. 9 shows the delay controller. A 4-b coarse delay control is achieved by a series of inverter delay selectors. Fine delay control is afforded by a 3-b-controlled MOS variable capacitor. T_d is chosen so that all PVT conditions can be covered; the coverage in typical condition is between 240 ps and 720 ps. This corresponds to 0.5 to 1.5 GHz. The longest delay also helps very slow conversions, because it can help to avoid charge leakage in the CDAC. The delay controller is composed of digital circuits that are smaller in area and consume less power than a phase locked loop (PLL).

IV. EXPERIMENTAL RESULTS

The prototype ADC was fabricated in a one-polysilicon seven-metal 65-nm CMOS process with MIM capacitors. A microphotograph is shown in Fig. 10. The ADC active area was 0.039 mm^2 . All digital calibration circuits were implemented on a chip measuring 0.013 mm^2 , which is 33% of the ADC area. If the CDAC linearity calibration were not implemented, CDAC would have to be designed using a six upper bits + four lower bits split CDAC design with 20-fF capacitors. The CDAC area will be twice as large as the current design, but it will not need a calibration scheme. So the difference is small from the viewpoint of area. However, the CDAC linearity calibration has the advantage of tolerance to process variations and input load.

Figs. 11 and 12 shows the effects of the calibration in the code density test. Before calibration, the peak DNL and INL were 1.4 LSB and 3.9 LSB, respectively; these values were mostly caused by the mismatch of the split capacitor C_{B1} . After calibration, peak DNL and INL were improved to 0.82 LSB and 0.72 LSB, respectively.

Fig. 13 illustrates the results of the dynamic performance measurement at 50 MSamples/s and a 1.0-V power supply. SNDR was 56.9 dB ($\text{ENOB} = 9.16$) and the spurious-free dynamic range (SFDR) was 75.2 dB at a 2-MHz input frequency, whereas the SNDR was 56.6 dB ($\text{ENOB} = 9.10$) and the SFDR was 76.9 dB under the Nyquist condition. The effective resolution bandwidth was 150 MHz.

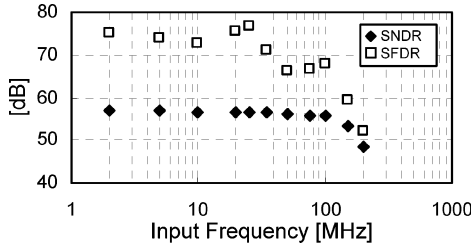


Fig. 13. Measured SNDR and SFDR versus the input frequency at 50 MSamples/s.

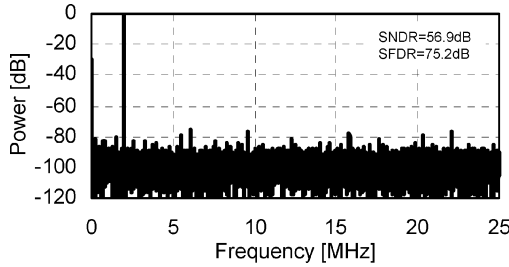


Fig. 14. Measured ADC output spectrum at 2-MHz input.

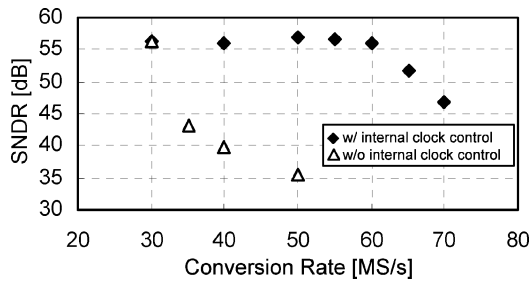


Fig. 15. Measured SNDR versus conversion rate at the 2-MHz input with and without internal clock control.

Fig. 14 shows the spectrum at the 2-MHz input. The SFDR was 75.2 dB and SNDR was 56.9 dB, which corresponds to an ENOB of 9.16. Fig. 15 plots the measured SNDR versus the conversion rate with and without the internal clock control. Without the internal clock control, the delay T_d was fixed at 30 MSamples/s and conversion speed increases. Lower bit comparison could not be completed due to the lack of comparison cycles. The internal clock control could track the conversion speed. The degradation at more than 70 MSamples/s was caused by incomplete settling of the CDAC. Fig. 16 shows the power-supply dependency with and without the internal clock control. Without the internal clock control, the ADJ code was fixed at a 1.3-V supply and the VDD was swept. As the supply fell, T_d became longer for the same ADJ code. This means that the lower bit comparison could not be achieved within the conversion phase. The internal clock control adjusted suitable T_d to the variation in the supply voltage. Fig. 17 shows the temperature dependency of SNDR and SFDR when the CDAC linearity calibration was performed at 25°C. The SNDR variation between -30°C to 120°C was less than 1 dB. Table I summarizes the performance of the ADC.

V. CONCLUSION

The SAR ADC with a split-CDAC and internal clock generator uses two calibration techniques to ensure stable operation.

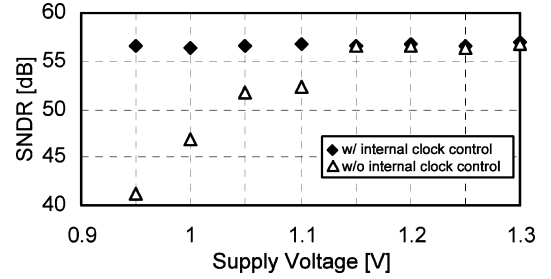


Fig. 16. Measured SNDR versus supply voltage at 50 MSamples/s and 2-MHz input with and without internal clock control.

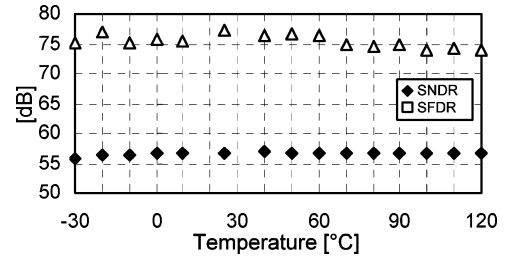


Fig. 17. Measured SNDR and SFDR versus temperature at 50 MSamples/s and 2-MHz input.

TABLE I
ADC PERFORMANCE SUMMARY

Technology	65 nm CMOS
Supply voltage	1.0 V
Resolution	10 bit
Conversion Rate	50MS/s
Input Range	1.0 V _{pp} differential
SNDR	56.9 dB@2MHz input 56.6 dB@25MHz input
SFDR	75.2 dB@2MHz input 76.9 dB@25MHz input
ENOB	9.16 @2MHz input 9.10 @25MHz input
DNL	< 0.82 LSB
INL	< 0.72 LSB
Power	820 μ W
Consumption	(Ana.: 450 μ W , Dig.: 370 μ W)
Active Area	0.039 mm ²

Digital calibration improves CDAC linearity, which is affected by parasitics and mismatching of the capacitors. The internal clock is controlled in the background to compensate for PVT variations. The calibration scheme is implemented on-chip, so no off-chip software processing is needed. The 10-b SAR ADC can perform 50 MSamples/s with a 1.0-V power supply. Power consumption is 820 μ W, including the power of the CDAC operation. This means that the ADC needs no additional power for providing the reference voltages. The active area, including the digital calibration circuits, is 0.039 mm².

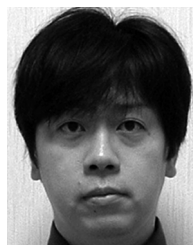
The FoM, defined as $\text{FoM} = P / (2^{\text{ENOB}} * 2\text{BW})$, is 29.7 fJ/conv-step. Table II compares state-of-the-art ADCs. If the power and area of the ADCs are especially small, more ADCs can be implemented in one chip. This makes it possible to build more compact portable medical equipment, such as ultrasound scanners.

TABLE II
COMPARISON OF STATE-OF-THE-ART ADCs

	Architecture	Tech.	Resolution [bit]	Conversion Rate [MS/s]	SNDR @Nyquist [dB]	Power [mW]	Area [mm ²]	FoM [fJ/conv-step]
[11] ISSCC' 10	SAR	65nm	10	100	56.0	1.13	0.026	21.9
This work	SAR	65nm	10	50	56.6	0.82	0.039	29.7
[5] ISSCC' 10	SAR	130nm	12	45	67.1	3.02	0.059	36.3
[12] VLSI' 09	SAR	130nm	10	50	52.8	0.92	0.075	51.6
[13] ISSCC' 08	Pipeline	65nm	10	100	59.0	4.50	0.070	61.8
[14] ISSCC' 07	SAR	90nm	9	50	48.7	0.73	0.080	65.6
[15] ISSCC' 08	SAR	90nm	9	40	51.3	0.82	0.090	68.3
[1] JSSC' 09	Pipeline	90nm	12	50	62.0	4.50	0.300	87.5
[16] ISSCC' 10	Pipeline+SAR	65nm	10	40	52.5	1.21	0.060	87.8

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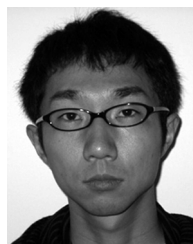
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